# LECTURE 15: REGRESSION DISCONTINUITY DESIGNS

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Economics 272, GSB 507, Spring 2025

# OUTLINE

# INTRODUCTION

- A Regression Discontinuity (RD) Design is a powerful and widely applicable identification strategy.
- Often access to, or incentives for participation in, a service or program is assigned based on transparent rules with criteria based on clear cutoff values, rather than on discretion of administrators.
- Comparisons of individuals that are similar but on different sides of the cutoff point can be credible estimates of causal effects for a specific subpopulation.
- Good for internal validity, not much external validity.
  Long history in Psychology literature (Thistlewaite and Campbell, 1960), early work by Goldberger (1972), recent resurgence in economics.

## BASICS

- Two potential outcomes $Y_i(\mathrm{C})$ and $Y_i(\mathrm{T})$, causal effect $Y_i(\mathrm{T}) - Y_i(\mathrm{C})$,
- binary treatment indicator $W_i$, covariate $X_i$, and the observed outcome:

$$Y_i \equiv Y_i\,(W_i) = \begin{cases} Y_i(\mathrm{C}) & \text{if } W_i = 0, \\ Y_i(\mathrm{T}) & \text{if } W_i = 1. \end{cases}$$

- At $X_i = c$ incentives to participate change.
- Two cases,
  - Sharp Regression Discontinuity:

$$W_i = \mathbf{1}_{X_i \geq c}$$

  - Fuzzy Regression Discontinuity Design:

$$\lim_{x \downarrow c} \Pr(W_i = 1 | X_i = x) \neq \lim_{x \uparrow c} \Pr(W_i = 1 | X_i = x)$$

## SHARP REGRESSION DISCONTINUITY

- Example (Lee, 2007)
- What is effect of incumbency on election outcomes? (More specifically, what is the probability of a Democrat winning the next election given that the last election was won by a Democrat?)
- Compare election outcomes in cases where previous election was very close.

# SHARP REGRESSION DISCONTINUITY DESIGNS

- Key assumption:
  $\mathbb{E}[Y(\mathrm{C})|X = x]$ and $\mathbb{E}[Y(\mathrm{T})|X = x]$ are continuous in $x$.
- Under this assumption,

$$\tau_{\text{srd}} = \lim_{x \downarrow c} \mathbb{E}[Y_i|X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i|X_i = x]$$

- The estimand (target) is the difference of two regression functions at a point.
- Extrapolation is unavoidable.

# FUZZY REGRESSION DISCONTINUITY

- Example VanderKlaauw (2002)
- What is effect of financial aid offer on acceptance of college admission.
- College admissions office puts applicants in a few categories based on numerical score.
- Financial aid offer is highly correlated with category.
- Compare individuals close to cutoff score.

# FUZZY REGRESSION DISCONTINUITY DESIGNS

- What do we look at in the FRD case: ratio of discontinuities in regression function of outcome and treatment:

$$\tau_{\text{frd}} = \frac{\lim_{x \downarrow c} \mathbb{E}[Y_i | X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i | X_i = x]}{\lim_{x \downarrow c} \mathbb{E}[W_i | X_i = x] - \lim_{x \uparrow c} \mathbb{E}[W_i | X_i = x]}$$

# INTERPRETATION OF FRD (HAHN, TODD, VANDERKLAAUW, 2001)

- Let $W_i(x)$ be potential treatment status given cutoff point $x$, for $x$ in some small neigborhood around $c$ (which requires that the cutoff point is at least in principle manipulable)
- $W_i(x)$ is non-increasing in $x$ at $x = c$.
- A complier is a unit such that

$$\lim_{x \downarrow X_i} W_i(x) = 0, \qquad \text{and} \lim_{x \uparrow X_i} W_i(x) = 1.$$

Then

$$\hat{\tau}^{\text{frd}} \equiv \frac{\lim_{x\downarrow c} \mathbb{E}[Y_i|X_i = x] - \lim_{x\uparrow c} \mathbb{E}[Y_i|X_i = x]}{\lim_{x\downarrow c} \mathbb{E}[W_i|X_i = x] - \lim_{x\uparrow c} \mathbb{E}[W_i|X_i = x]}$$

$$= \mathbb{E}[Y_i(\mathrm{T}) - Y_i(\mathrm{C})|\text{unit } i \text{ is a complier and } X_i = c].$$

# EXTERNAL VALIDITY

- The estimatand has little external validity.
- It is at best valid for a population defined by
  - the cutoff value $c$,
  - the complier subpopulation that is affected at that value.

# FUZZY REGRESSION DISCONTINUITY DESIGNS VERSUS UNCONFOUNDEDNESS

- Unconfoundedness:

$$W_i \perp\!\!\!\perp Y_i(\mathrm{C}), Y_i(\mathrm{T}) \,\Big|\, X_i.$$

$$\text{so } \mathbb{E}[Y_i(\mathrm{T}) - Y_i(\mathrm{C}) | X_i = c] = \mathbb{E}[Y_i | W_i = 1, X_i = c] - \mathbb{E}[Y_i | W_i = 0, X_i = c].$$

- This approach assumes that differences between treated and control units with $X_i = c$ have a causal interpretation, without exploiting the discontinuity.
- Unconfoundedness assumes units are comparable if their covariates are similar, even if not identical.
- This is not an attractive assumption in the current setting where the probability of receiving the treatment is discontinuous in the covariate.
- Even if unconfoundedness holds, under continuity of potential outcome means FRD approach will be consistent for the average effect for compliers at $X_i = c$.

## GRAPHICAL ANALYSES

- Three key plots:

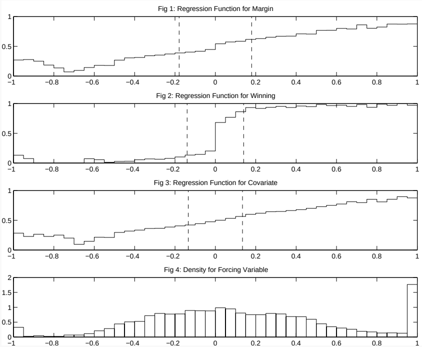
  **A.** Plot regression function $\mathbb{E}[Y_i|X_i = x]$

  **B.** Plot regression functions $\mathbb{E}[Z_i|X_i = x]$ for covariates that do not enter the assignment rule $Z_i$

  **C.** Plot density $f_X(x)$.

- In all cases use estimators that do not smooth around the cutoff value.
- For example, for binwidth $h$ define bins $[b_{k-1}, b_k]$, where $b_k = c - (K_0 - k + 1) \cdot h$, and average outcomes within bins.

# GRAPHICAL ANALYSES



# ESTIMATION: PITFALLS OF LOCAL CONSTANT REGRESSION

- We are interested in the value of a regression function at the boundary of the support. Standard kernel regression

$$\widehat{\mu_l(c)} = \sum_{i|c-h<X_i<c}^{N} Y_i \Bigg/ \sum_{i|c-h<X_i<c}^{N} 1 \tag{1}$$

does not work well for that case (slow convergence rates), and is not widely used here.

## ESTIMATION: PITFALLS OF LOCAL CONSTANT REGRESSION

- Suppose we estimate a regression function at an **interior** point

$$E[Y_i|X_i = x] = \mu(x)$$

$$\widehat{\mu(c)} = \sum_{i|c-h<X_i<c+h}^{N} Y_i \Bigg/ \sum_{i|c-h<X_i<c+h}^{N} 1$$

- Then

$$E\left[\widehat{\mu(c)}|X_1, \ldots X_N\right] = \sum_{i|c-h<X_i<c+h}^{N} \mu(X_i) \Bigg/ \sum_{i|c-h<X_i<c+h}^{N} 1 \approx \mu(c) + c_1 h^2$$

- This is because the errors on the left and the right cancel each other out to the first order.

# ESTIMATION: PITFALLS OF LOCAL CONSTANT REGRESSION

- When we are interested in the regression function at the boundary, the convergence rate is slower:

$$\widehat{\mu_l(c)} = \sum_{i|c-h<X_i<c}^{N} Y_i \Big/ \sum_{i|c-h<X_i<c}^{N} 1$$

- Now

$$E\left[\widehat{\mu(c)}|X_1, \ldots X_N\right] \approx \mu_l(c) + c_1 h$$

- Bias is proportional to $h$ for a boundary point instead of to $h^2$ as it is for an interior point.

# ESTIMATION: PITFALLS OF GLOBAL POLYNOMIAL REGRESSION

- More popular, but not recommended, is to use a higher order polynomial approximation.
- This tends to be sensitive to the order of the polynomial used, and implicitly give substantial weight to observations far away from the threshold – same problem as with Lalonde data.

# ESTIMATION: LOCAL LINEAR REGRESSION

- Better rates are obtained by using local linear regression. First

$$\min_{\alpha_l, \beta_l} \sum_{i|c-h<X_i<c}^{N} \left(Y_i - \alpha_l - \beta_l \cdot (X_i - c)\right)^2,$$

- The value of left hand limit $\mu_l(c)$ is then estimated as

$$\widehat{\mu_l(c)} = \hat{\alpha}_l + \hat{\beta}_l \cdot (c - c) = \hat{\alpha}_l.$$

- Faster convergence rate than local constant / kernel regression.

## ESTIMATION: LOCAL LINEAR REGRESSION

- Alternatively one can estimate the average effect directly in a single regression,

$$Y_i = \alpha + \beta \cdot (X_i - c) + \tau \cdot W_i + \gamma \cdot (X_i - c) \cdot W_i + \varepsilon_i$$

- This si equivalent to solving

$$\min_{\alpha,\beta,\tau,\gamma} \sum_{i=1}^{N} \mathbf{1}_{c-h \leq X_i \leq c+h}$$

$$\times \left(Y_i - \alpha - \beta \cdot (X_i - c) - \tau \cdot W_i - \gamma \cdot (X_i - c) \cdot W_i\right)^2,$$

which will numerically yield the same estimate of $\tau_{\text{srd}}$.

- This interpretation extends easily to the inclusion of covariates.

## ESTIMATION FOR THE FUZZY REGRESSION DISCONTINUITY CASE

- Do local linear regression for both the outcome and the treatment indicator, on both sides,

$$\left(\hat{\alpha}_{yl}, \hat{\beta}_{yl}\right) = \arg \min_{\alpha_{yl}, \beta_{yl}} \sum_{i: c-h \leq X_i < c} \left(Y_i - \alpha_{yl} - \beta_{yl} \cdot (X_i - c)\right)^2,$$

$$\left(\hat{\alpha}_{wl}, \hat{\beta}_{wl}\right) = \arg \min_{\alpha_{wl}, \beta_{wl}} \sum_{i: c-h \leq X_i < c} \left(W_i - \alpha_{wl} - \beta_{wl} \cdot (X_i - c)\right)^2,$$

- Similarly $(\hat{\alpha}_{yr}, \hat{\beta}_{yr})$ and $(\hat{\alpha}_{wr}, \hat{\beta}_{wr})$.
- Then the FRD estimator is

$$\hat{\tau}_{\text{frd}} = \frac{\hat{\tau}_y}{\hat{\tau}_w} = \frac{\hat{\alpha}_{yr} - \hat{\alpha}_{yl}}{\hat{\alpha}_{wr} - \hat{\alpha}_{wl}}.$$

## ESTIMATION FOR THE FUZZY REGRESSION DISCONTINUITY CASE

- Alternatively, define the vector of covariates

$$V_i = \begin{pmatrix} 1 \\ \mathbf{1}_{X_i<c} \cdot (X_i - c) \\ \mathbf{1}_{X_i \geq c} \cdot (X_i - c) \end{pmatrix}, \qquad \text{and } \delta = \begin{pmatrix} \alpha_{yl} \\ \beta_{yl} \\ \beta_{yr} \end{pmatrix}.$$

- Then we can write

$$Y_i = \delta' V_i + \tau \cdot W_i + \varepsilon_i.$$

- Then estimating $\tau$ by TSLS, using
  - $W_i$ as the endogenous regressor,
  - the indicator $\mathbf{1}_{X_i \geq c}$ as the excluded instrument
  - $V_i$ as the set of exogenous variables
- This is is numerically identical to $\hat{\tau}_{\text{frd}}$ before (because of uniform kernel)

# ESTIMATION: LOCAL QUADRATIC REGRESSION

- Local quadratic regression can remove some bias

$$\min_{\alpha_l, \beta_l} \sum_{i|c-h<X_i<c}^{N} \left( Y_i - \alpha_l - \beta_l \cdot (X_i - c) - \gamma_l \cdot (X_i - c)^2 \right)^2,$$

- The value of lefthand limit $\mu_l(c)$ is then estimated as

$$\widehat{\mu_l(c)} = \hat{\alpha}_l + \hat{\beta}_l \cdot (c - c) + \hat{\gamma}_l \cdot (c - c)^2 = \hat{\alpha}_l.$$

## CHOOSING THE BANDWIDTH (IMBENS-KALYANARAMAN, 2012)

- We wish to take into account that
  - we are interested in the regression function at the boundary of the support,
  - we are interested in the regression function at $x = c$.
- In standard nonparametric regression we can use a hold out (test) sample, and choose the bandwidth the minimize the average squared error in the test sample.
- That is not feasible here: we have no unbiased estimator for the true value of the regression function at the boundary point.

# CHOOSING THE BANDWIDTH (IMBENS-KALYANARAMAN, 2012)

- IK focus on minimizing

$$\mathbb{E}\left[\left(\hat{\mu}_l(c) - \mu_l(c)\right)^2 + (\hat{\mu}_r(c) - \mu_r(c))^2\right]$$

- Both $\hat{\mu}_l(c)$ and $\hat{\mu}_r(c)$ are based on local linear estimators, with the same bandwidth $h$.
- Assume continuous density for forcing variable $X_i$.
- Assume true regression function is quadratic.

# OPTIMAL BANDWIDTH

- Result:

$$h_{\text{opt}} = \left(\frac{C_2}{C_1}\right)^{1/5} \cdot \left( \frac{\frac{\sigma_r^2(c)}{p \cdot f_r(c)} + \frac{\sigma_l^2(c)}{(1-p) \cdot f_l(c)}}{\left(\frac{\partial^2 m_r}{\partial x^2}(c)\right)^2 + \left(\frac{\partial^2 m_l}{\partial x^2}(c)\right)^2} \right)^{1/5} \cdot N^{-1/5}$$

- $p$ is share of observations above threshold.

$$C_1 = \frac{1}{4} \cdot \left(\frac{\nu_2^2 - \nu_1\nu_3}{\nu_2\nu_0 - \nu_1^2}\right)^2 \qquad C_2 = \frac{\int_0^\infty (\nu_2 - u\nu_1))^2 K^2(u)du}{\left(\nu_2\nu_0 - \nu_1^2\right)^2}$$

$$\nu_j = \int_0^\infty u^j K(u)du$$

- If $K(u) = 1_{|u|<0.5}$, then $(C_2/C_1) = 5.40$

# BANDWIDTH FOR FRD DESIGN

- Two approaches:
  - Calculate optimal bandwidth separately for both regression functions and choose smallest.
  - Calculate optimal bandwidth only for outcome and use that for both regression functions.
- Typically the regression function for the treatment indicator is flatter than the regression function for the outcome away from the discontinuity point (completely flat in the SRD case). So using same criterion would lead to larger bandwidth for estimation of regression function for treatment indicator.
- In practice it is easier to use the same bandwidth, and so to avoid bias, use the bandwidth from criterion for SRD design or smallest.

# ALTERNATIVE BANDWIDTH SELECTION METHODS

- Calonico, Cattaneo & Titiunik (2014) note that using the IK bandwidth because it balances variance and bias-squared) leads to a bias in the estimator that invalidates normal-distribution-based confidence intervals.
- They propose using the optimal bandwith based on a local linear estimator, but then using a local quadratic estimator. This removes the leading term in the bias, and so regular confidence intervals work.

# ALTERNATIVE WEIGHT SELECTION

- Recent interesting work by Armstrong and Kolesár (2018), also Imbens and Wager (2018).
- Sharp RD Define

$$\mu_0(x) = \mathbb{E}[Y_i | X_i = x, x \leq c], \qquad \mu_1(x) = \mathbb{E}[Y_i | X_i = x, x > c].$$

- Observations $(X_i, Y_i)$, $i = 1, \ldots, N$.
- Define weights $\omega_i$, $i = 1, \ldots, N$, $\sum_{i:X_i > c} \omega_i = \sum_{i:X_i \leq c} \omega_i = 1$.
- Estimator

$$\hat{\tau}(\omega) = \sum_{i:X_i > c} \omega_i Y_i - \sum_{i:X_i < c} \omega_i Y_i.$$

## ALTERNATIVE WEIGHT SELECTION

- For any choice of $\mu_W(\cdot)$ and any set of weights we can figure out the bias in the estimator $\hat{\tau}(\omega)$ and construct valid confidence intervals.
- The idea in Armstrong and Kolesár (2018) and IW is to choose $\omega$ to minimize squared error, over a space of functions. Choosing the $\omega_i$ to put all the weight close to $c$ leads to small maximal bias, but large variance. Spreading the weights out leads to large maximal bias, but small variance.
- AK restrict the set to satisfy Lipschitz conditions with some Lipschitz constant. This leads to analytic solutions for $\omega$.
- Imbens and Wager (2018) restrict the set by assuming smoothness but restricting the second derivatives of $\mu_W(\cdot)$. This requires numerical solutions.

# SIMULATION STUDY BASED ON GAN DESIGNS

| | `rdrobust` (rob.) | | `RDHonest` | | `plrd` | |
|---|---|---|---|---|---|---|
| **Dataset** | **EC** | **width** | **EC** | **width** | **EC** | **width** |
| lee2008 | 95.4% | 0.042 | 97.6% | 0.052 | 98.3% | 0.035 |
| cattaneo2015senate | 95.1% | 6.281 | 97.4% | 6.980 | 97.3% | 4.525 |
| matsudaira (read.) | 96.9% | 0.069 | 98.4% | 0.096 | 97.1% | 0.060 |
| matsudaira (math) | 93.0% | 0.054 | 97.8% | 0.072 | 97.0% | 0.051 |
| jacob2004 (read.) | 95.0% | 0.080 | 98.2% | 0.104 | 96.5% | 0.065 |
| jacob2004 (math) | 93.4% | 0.093 | 97.1% | 0.120 | 95.0% | 0.069 |
| Oreopoulos | 94.6% | 0.264 | 98.9% | 0.316 | 96.1% | 0.194 |
| LudwigMiller | 95.7% | 4.786 | 97.6% | 5.238 | 97.4% | 3.542 |
| meyersson2014islamic | 93.3% | 7.531 | 96.3% | 8.099 | 97.3% | 5.095 |

## VARIANCE ESTIMATION

- Define

$$\sigma^2_{Yl} = \lim_{x \uparrow c} \mathrm{Var}(Y_i | X_i = x), \qquad C_{YWl} = \lim_{x \uparrow c} \mathrm{Cov}(Y_i, W_i | X_i = x),$$

$$V_{\tau_y} = \frac{4}{f_X(c)} \cdot \left( \sigma^2_{Yr} + \sigma^2_{Yl} \right), \qquad V_{\tau_w} = \frac{4}{f_X(c)} \cdot \left( \sigma^2_{Wr} + \sigma^2_{Wl} \right)$$

$$C_{\tau_y, \tau_w} = Nh\mathrm{covar}\left( \hat{\tau}_y, \hat{\tau}_w \right) = \frac{4}{f_X(c)} \cdot \left( C_{YWr} + C_{YWl} \right).$$

- The asymptotic distribution has the form

$$\sqrt{Nh} \cdot (\hat{\tau} - \tau) \xrightarrow{d} \mathcal{N}\left( 0, \frac{1}{\tau_w^2} \cdot V_{\tau_y} + \frac{\tau_y^2}{\tau_w^4} \cdot V_{\tau_w} - 2 \cdot \frac{\tau_y}{\tau_w^3} \cdot C_{\tau_y, \tau_w} \right).$$

- Special case of that in HTV, using the rectangular kernel, and with $h = N^{-\delta}$, for $1/5 < \delta < 2/5$ (so asymptotic bias can be ignored).

# TSLS VARIANCE FOR FRD DESIGN

- The second estimator for the asymptotic variance of $\hat{\tau}$ exploits the interpretation of the $\hat{\tau}$ as a TSLS estimator.
- The variance estimator is equal to the robust variance for TSLS based on
  - the subsample of observations with $c - h \leq X_i \leq c + h$,
  - using the indicator $\mathbf{1}_{X_i \geq c}$ as the excluded instrument,
  - the treatment $W_i$ as the endogenous regressor
  - and the $V_i$ as the exogenous covariates.

## CONCERNS ABOUT VALIDITY

- Two main conceptual concerns in the application of RD designs, sharp or fuzzy.
  - Other Changes Possibility of other changes at the same cutoff value of the covariate. Such changes may affect the outcome, and these effects may be attributed erroneously to the treatment of interest.
  - Manipulation of Forcing Variable The second concern is that of manipulation of the covariate value.

# SPECIFICATION CHECKS

- What type of specification checks can we do?
  - Discontinuities in Average Covariates
  - A Discontinuity in the Distribution of the Forcing Variable
  - Discontinuities in Average Outcomes at Other Values
  - Sensitivity to Bandwidth Choice

## DISCONTINUITIES IN AVERAGE COVARIATES

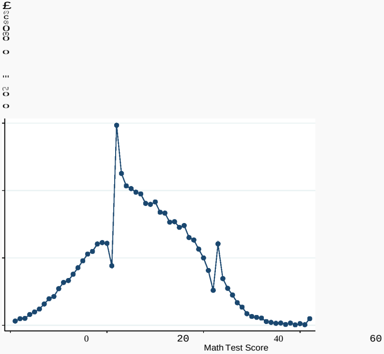
- Test the null hypothesis of a zero average effect on pseudo outcomes known not to be affected by the treatment.
- Such variables includes covariates that are by definition not affected by the treatment. Such tests are familiar from settings with identification based on unconfoundedness assumptions.
- Although not required for the validity of the design, in most cases, the reason for the discontinuity in the probability of the treatment does not suggest a discontinuity in the average value of covariates.
- If we find such a discontinuity, it typically casts doubt on the assumptions underlying the RD design.

# A DISCONTINUITY IN THE DISTRIBUTION OF THE FORCING VARIABLE

- McCrary (2007) suggests testing the null hypothesis of continuity of the density of the covariate that underlies the assignment at the discontinuity point, against the alternative of a jump in the density function at that point.
- Again, in principle, the design does not require continuity of the density of $X$ at $c$, but a discontinuity is suggestive of violations of the no-manipulation assumption.
- If in fact individuals partly manage to manipulate the value of $X$ in order to be on one side of the boundary rather than the other, one might expect to see a discontinuity in this density at the discontinuity point.

# DISCONTINUITY IN DENSITY OF RUNNING VARIABLE

- Diamond & Persson (2016)



# DISCONTINUITIES IN AVAREAGE OUTCOMES AT OTHER VALUES

- Taking the subsample with $X_i < c$ we can test for a jump in the conditional mean of the outcome at the median of the forcing variable.
- To implement the test, use the same method for selecting the binwidth as before. Also estimate the standard errors of the jump and use this to test the hypothesis of a zero jump.
- Repeat this using the subsample to the right of the cutoff point with $X_i \geq c$. Now estimate the jump in the regression function and at $q_{X,1/2,r}$, and test whether it is equal to zero.

## SENSITIVITY TO BANDWIDTH CHOICE

- One should investigate the sensitivity of the inferences to this choice, for example, by including results for bandwidths twice (or four times) and half (or a quarter of) the size of the originally chosen bandwidth.
- Obviously, such bandwidth choices affect both estimates and standard errors, but if the results are critically dependent on a particular bandwidth choice, they are clearly less credible than if they are robust to such variation in bandwidths.

# ILLUSTRATION BASED ON DAVID LEE ELECTION DATA

- Forcing variable is difference in dem vs rep vote share in last election.
- Outcomes are dem vote share in next election, and indicator for democrats winning the next election.
- Covariate for testing is dem vote share in prior election
- 6558 congressional elections.
- Uniform kernel with support $[-0.5, 0.5]$

# RESULTS FOR LEE DATA

| Outcome | IK Bandwidth | Estimate | (s.e.) |
|---|---|---|---|
| Dem Win Next Elect | 0.36 | 0.082 | (0.010) |
| Demt Margin Next Election | 0.27 | 0.412 | (0.039) |
| Dem Margin Prev Election | 0.28 | -0.003 | (0.013) |

# SIMULATIONS BASED ON GENERATIVE ADVERSERIAL NETWORKS USING LEE DATA

| Method | average est | mse | bias | sd |
|---|---|---|---|---|
| Local Linear IK bandwidth | 0.0706 | 0.0096 | -0.0005 | 0.0096 |
| Quadratic | 0.0721 | 0.0152 | 0.0010 | 0.0152 |
| Armstrong-Kolesar | 0.0724 | 0.0171 | 0.0013 | 0.0171 |
| Imbens-Wager | 0.0722 | 0.0170 | 0.0012 | 0.0170 |

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